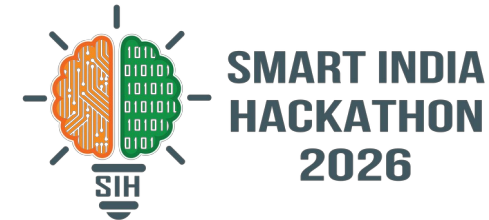


SMART INDIA HACKATHON 2026



CASSANDRA AI

- **Problem Statement ID** – SIH26184
- **Problem Statement Title** – Predictive Analytics Framework for Cybercrime Complaints to Forecast Likely Cash-Withdrawal Locations
- **Theme** – Blockchain & Cybersecurity
- **PS Category** – Software
- **Team ID** – 187045
- **Team Name** – NeferX

CASSANDRA: Cash-out Anomaly Surveillance & Spatial-temporal Analytics for Network Detection and Risk Assessment



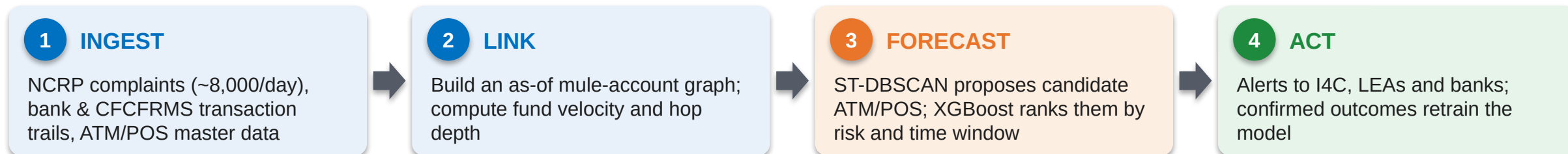


CASSANDRA AI

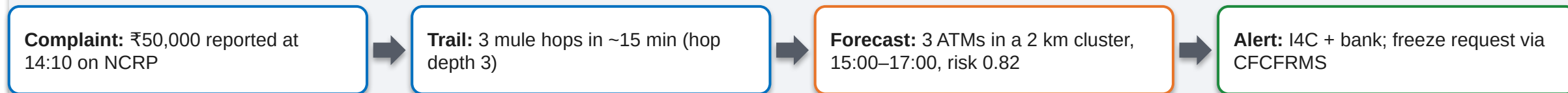
❖ Proposed Solution

Cash-out Anomaly Surveillance & Spatial-temporal Analytics for Network Detection and Risk Assessment

Detailed explanation: predict where and when stolen money will be cashed out



Worked example (illustrative numbers)



How it addresses the problem

- **Reactive** → **proactive**: targets the gap between mule credit and cash-out; lead time is measured on backtest, not assumed
- **Where + when**: ranked ATM/POS with a 2-hour window and risk score
- **All 4 PS deliverables**: Engine, Heatmap, LE Interface, Alerts routed I4C → LEA → bank

Innovation and uniqueness

- **Signal fusion**: complaints + mule graph + fund velocity + geospatial history, not hotspot maps alone
- **Trail-aware**: ranks terminals near the trail's latest hop, not just past crime spots
- **Explainable + learning**: SHAP shows why an alert fired; confirmed outcomes retrain the model

❖ Technologies to be used

Data & Streaming

- Python, Pandas, Prefect
- pgmq queue (Kafka later)
- H3 spatial index

ML & Geospatial

- XGBoost + SHAP
- ST-DBSCAN, Splink
- NetworkX (mule graph)

Storage & API

- PostgreSQL + PostGIS
- FastAPI (REST hooks)
- Docker

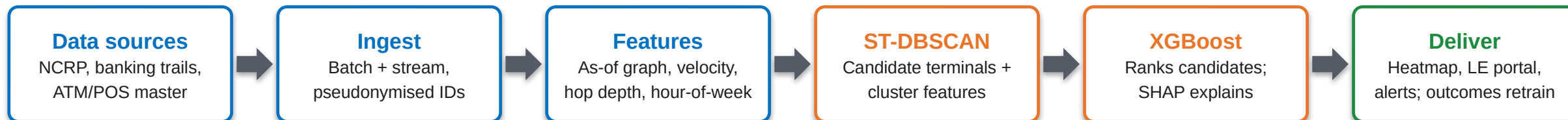
UI & Alerts

- React LE portal + map
- Grafana (demo view)
- MSG91 SMS, email

MLOps & Security

- MLflow, Prometheus
- RBAC + audit logs
- Encryption (DPDP)

❖ Methodology and process for implementation



ML formulation

- **Unit:** (ATM/POS, 2-hour window)
- **Label:** 1 if a complaint-linked mule withdraws there in that window

Leakage control and baseline

- **Split:** rolling-origin, 7-day embargo, no future features
- **Baseline to beat:** past-hotspot frequency per terminal

Data for the prototype (sources and links: References D1–D8)

Calibrate (real, public)

NCRP state totals (D5), RBI ATM counts (D6), real ATM coordinates from OSM (D7)

Benchmark modules (real, public)

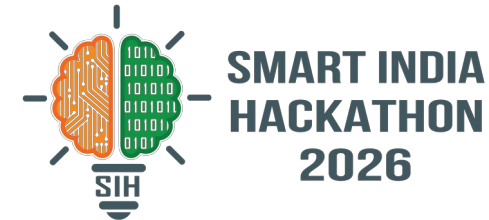
Mule graph: D1, D2 · timing: D3 · imbalance and drift: D4 · ST-DBSCAN: D8

Generate (simulator)

Calibrated agent-based simulator builds complaint → mule chain → cash-out; an LLM only writes scenarios and complaint text



FEASIBILITY AND VIABILITY



❖ Analysis of the feasibility of the idea

Technical

MED-HIGH

Mature open-source parts; the real risk is data linkage and labels, not tooling

Operational

MEDIUM

Fits I4C / CFCFRMS workflow; needs SOPs for alert routing and accountability

Data access

KEY RISK

Needs bank and NCRP data-sharing agreements and DPDP-compliant handling

Cost

LOW-MED

No licence fees; infrastructure and integration effort to be costed in the pilot

❖ Potential challenges and risks

No dataset; labels and bank data access

False alerts waste field resources

Privacy and legal exposure (DPDP Act 2023)

Model drift and biased feedback labels

Alert arrives after the withdrawal

❖ Strategies for overcoming

Calibrated simulator (D1–D8) for the prototype; standard API/CSV adapters so anonymised I4C and bank feeds plug into a pilot

Alert above a tuned threshold; show SHAP top factors; officer confirms; named owner per alert; track precision@k

Data minimisation, pseudonymised IDs, role-based access, audit logs, encryption in transit and at rest

MLflow drift monitoring, scheduled retraining, sample unconfirmed alerts to reduce label bias

Measure complaint-to-alert latency; target the credit-to-cash-out gap; auto freeze request via CFCFRMS

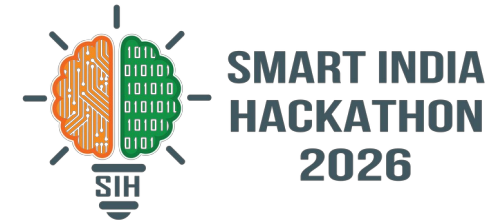
Phase 1 · Prototype: synthetic data, backtest, dashboard demo

Phase 2 · Pilot: one state + one bank, anonymised feeds via I4C

Phase 3 · Scale: multi-state rollout, CFCFRMS integration



IMPACT AND BENEFITS



❖ Potential impact on the target audience

LEAs & I4C

Know which ATMs and time windows to cover, so field teams are deployed with intent instead of after the fact.

Banks & FIs

Earlier warning to hold funds and alert branches and ATMs, so fewer rupees leave the system.

Citizen victims

Better chance of recovery, faster response and more trust in reporting on NCRP.

❖ Benefits of the solution

Social

Fewer citizens lose savings; stronger public confidence in reporting cybercrime

Economic

Higher fund-blocking and recovery rates; lower fraud losses for banks

Operational

Prioritised alerts replace manual triage across a rising complaint volume

National security

Cross-state syndicate visibility through connected transaction trails

PILOT DESIGN GOALS: hypotheses to calibrate on real data; synthetic results only validate the pipeline

> 1×

lift over the past-hotspot baseline, reported with confidence intervals

< 60 s

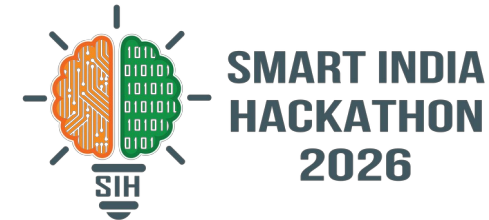
complaint-to-alert processing latency (system design goal)

Measured

precision@10 and median alert lead time, reported on backtest; targets set after the pilot



RESEARCH AND REFERENCES



❖ Details / Links of the reference and research work

Datasets used to build the synthetic data (D1–D8)

D1 IBM AMLworld (Altman et al., NeurIPS 2023): agent-based synthetic bank transactions, 5M–180M rows, complete laundering labels. [Kaggle](#) [arXiv:2306.16424](#)

D2 Elliptic / Elliptic++ (Weber et al. 2019; Elmougy & Liu, KDD 2023): labelled Bitcoin transaction graph, 203,769 nodes, ~2% illicit. [Kaggle](#) [Elliptic++](#) [arXiv:1908.02591](#)

D3 PaySim (Lopez-Rojas et al., 2016): mobile-money simulator calibrated on real logs; cash-out timing and velocity. [Kaggle](#) [GitHub](#)

D4 Bank Account Fraud (Jesus et al., NeurIPS 2022): privacy-preserving bank fraud data with imbalance and temporal drift. [Kaggle](#)

D5 NCRP state/UT-wise statistics (OGD India): incidents, amount reported, lien amount. [data.gov.in](#)

D6 RBI ATM deployment state and region-wise, quarterly. [rbi.org.in](#)

D7 OpenStreetMap ATMs (amenity=atm, ODbL): real ATM coordinates and operators. [OSM wiki](#)

D8 Chicago Crimes 2001–present (city open data): geotagged, time-stamped events; used only to check the ST-DBSCAN pipeline. [data.cityofchicago.org](#)

Research and methods

Butt et al. (2020), Spatio-Temporal Crime HotSpot Detection and Prediction, IEEE Access 8. [doi:10.1109/ACCESS.2020.3022680](#)

Birant & Kut (2007), ST-DBSCAN, Data & Knowledge Engineering 60(1). [doi:10.1016/j.datak.2006.01.013](#)

Chen & Guestrin (2016), XGBoost, KDD. [doi:10.1145/2939672.2939785](#)

Borisov et al. (2023), Language Models are Realistic Tabular Data Generators, ICLR. [arXiv:2210.06280](#)

National Cybercrime Reporting Portal (NCRP), MHA. [cybercrime.gov.in](#)

Policy, standards and platform

Digital Personal Data Protection Act, 2023. [meity.gov.in \(PDF\)](#)

IndiaAI Responsible AI Resource Kit. [indiaai.gov.in](#)

NIST AI Risk Management Framework 1.0. [nist.gov](#)

ISO/IEC 27001:2022. [iso.org](#)

OWASP ASVS 4.0.3. [owasp.org](#)